

Software Engineering Lab

UE24CS341A

Lab 3: Component Modelling & Architectural Pattern Selection

PROBLEM STATEMENT 6: ALUMNI MENTORSHIP & MOCK INTERVIEW PLATFORM

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1. Architecture Selection

We chose **Layered Architecture** for the Alumni Mentorship & Mock Interview Platform.

The system is organized into three logical layers:

- **Presentation Layer** – User Interface
- **Business Layer** – Mentor Matching, Availability & Booking, Interview Assessment, Experience Sharing
- **Data Layer** – Platform Database

2. Architecture Choice

The platform has a clear separation between user interaction, application/business functionality, and persistent data. A layered structure matches this organization and keeps the major responsibilities of the system separated.

i. **Clear Separation of Responsibilities:**

The platform contains different functionality such as mentor recommendation, session booking, interview assessment, and experience sharing. Layered architecture separates the user interface, business logic, and data management, making the system easier to understand and maintain.

ii. **Suitable for the Platform Workflow:**

The platform operations follow a clear flow from the user interface to business components and then to the database. For example, a student can request mentor recommendations through the interface, the Mentor Matching component processes the request, and the required mentor information is retrieved from the database.

3. Security Advantage

Layered architecture separates access to the database from the user interface. Users interact with the system through business components rather than directly accessing stored data. This helps protect user profiles, interview scorecards, and mentor contact information according to the privacy requirements that were identified in Lab 1.

4. Performance Benefit

The layered organization keeps application processing separate from direct data access. Components perform their required operations through defined interfaces, helping keep data access organized and supporting reasonable response time.

5. Conclusion

Layered Architecture provides a simple and structured architecture for the Alumni Mentorship & Mock Interview Platform while maintaining separation of responsibilities, data protection, and organized system operations.